MIL-E-50008B(EA)
15 April 1982
SUPERSEDING
MIL-E-50008A
30 March 1960

MILITARY SPECIFICATION

2-ETHYLHEXANOIC ACID, TECHNICAL

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of 2-ethylhexanoic acid.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

HH-I-585	-	Insulation, Thermal, (Vermiculite)
NN-P-71	-	Pallets, Material Handling Wood, Stringer Construction 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals
PPP-T-60	-	Tape: Packaging, Waterproof

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1168	-	Ammunition Lot Numbering

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D891	-	Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
D1078	-	Distillation Range of Volatile Organic Liquids
D1193	-	Reagent Water
D1209	-	Color of Clear Liquids (Platinum - Cobalt Scale)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical and physical properties. The 2-ethylhexanoic acid shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Specific gravity, 20°/20°C	0.904	0.909	4.2.4.1
Color (Pt - Co scale)	No darker than 50	No darker than 50	4.2.4.2
Distillation at 760 mm:			4.2.4.3
Initial boiling point, °C	222	---	
At 5 ml distilled, °C	224	---	
At 95 ml distilled, °C	---	230	
Dry point, °C	---	233	
Purity, percent by weight	95	---	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the 2-ethylhexanoic acid produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 300 milliliters (ml) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Container liner not as specified or missing
(g) Container pads missing (where required)
(h) Unitization not as specified

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 600 ml. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Specific gravity. Determine the specific gravity of the specimen using a calibrated pycnometer in accordance with ASTM D891 in a water bath maintained at 20.00° $\pm$ 0.05°C.

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4.2.4.2 Color. Determine the color of the specimen in accordance with ASTM D1209.

4.2.4.3 Distillation. Distill the specimen in accordance with ASTM D1078 using a calibrated thermometer having a range of 95° to 255°C with 0.5°C subdivisions. Conduct the distillation at 760 millimeters (mm) of mercury or correct by adding or subtracting 0.046°C for every mm under or above 760 mm.

4.2.4.4 Purity. Add 75 ml of methanol to a 250-ml wide-mouth Erlenmeyer flask. Add 2 drops of phenolphthalein indicator and adjust to a faint pink end point with 0.1N sodium hydroxide. Pipet 2 ml of specimen into a tared weighing bottle and weigh to an accuracy of ± 0.0002 gram. Place the weighing bottle in the flask and shake with a swirling motion until the sample and methanol are mixed. Titrate back to the same pink end point with 0.5N sodium hydroxide. Calculate the percent purity as follows:

$$\text{2-ethylhexanoic acid, percent by weight} = \frac{14.42\ AN}{W}$$

where: A = Milliliters of sodium hydroxide used,
N = Normality of the sodium hydroxide, and
W = Weight of the specimen in grams.

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A. A quantity of 1 US gallon (+1-1/2 or -0 fluid ounces) of 2-ethylhexanoic acid shall be unit packed in a glass bottle of nominal 1-gallon capacity. The bottle shall conform to PPP-C-186 group A, class 1, type D glass, style 1, light penetration grade 2, having closure B, and outer seal A. The liner shall be faced with a phenol-formaldehyde film or material of equal or greater resistance to 2-ethylhexanoic acid. The closure of the can or bottle, as applicable, shall be closed to a torque within a range as recommended by the manufacturer. Each bottle shall be overpacked, closure uppermost, in a fiberboard box having top and bottom pads. The box shall conform to PPP-B-636, grade V3c. Vermiculite conforming to HH-I-585, class 1, types II or III shall be used to form a 3-inch cushion between the sides of the bottle and inner sides of the box, and a 4-1/2-inch cushion between the top and bottom of the bottle and top and bottom of the box. All seams of the box shall be completely sealed by nominal 3-inch wide tape conforming to PPP-T-60, class optional.

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5.1.2 Level C. A quantity of 2-ethylhexanoic acid as specified (see 6.2), shall be unit packed commercially in a manner to assure purity of the acid and integrity of the container from supply source to destination, and for a minimum period of 1 year from date of pack. Unit packing shall be in conformity with all regulatory requirements, shall assure acceptance by common carrier, and shall be in accordance with Uniform Freight Classification and National Motor Freight Classification Rules.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four 1-gallon bottles of 2-ethylhexanoic acid, unit packed as specified in 5.1, shall be packed, closure end uppermost, in a close-fitting wood box. The box shall conform to PPP-B-585, class 3, with veneer face boards, style optional, for a type 2 average load. Each box shall be furnished with half-slotted fiberboard separators forming a close-fitting full-can-height cell for each can. Top and bottom pads, plus an inner liner, shall be inserted in each box. Additional fiberboard pads shall be inserted in the box where needed to prevent motion of contents. The box shall be closed in accordance with PPP-B-585.

5.2.2 Level B. 2-Ethylhexanoic acid shall be packed level B as specified for level A in 5.2.1 except that the box shall conform to PPP-B-636, grade V3c, for a type 2 average load.

5.2.3 Level C. 2-Ethylhexanoic acid, unit packed as specified in 5.1, shall be packed to assure carrier acceptance, safe delivery at destination, and conformity to Uniform Freight Classification and National Motor Freight Classification Rules.

5.3 Unitization. Level A and B packs of 2-ethylhexanoic acid shall be palletized, with closure ends of the bottles uppermost, in accordance with load type I of MIL-STD-147 using a pallet conforming to NN-P-71, type IV. Level C packs shall be unitized to assure carrier acceptance and safe arrival at destination.

5.4 Marking. Unit packs, packs, and unitized loads shall be marked in accordance with MIL-STD-129. The lot number and date of manufacture of the 2-ethylhexanoic acid shall be shown on each unit pack, pack, and pallet load. In addition, each unit pack, pack, and pallet load shall bear the following warning statement:

WARNING!

CAUSES IRRITATION

Avoid contact with eyes, skin, and clothing. Wash thoroughly after handling.

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FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician. Flush skin with water. (Wash clothing before reuse.)

6. NOTES

6.1 Intended use. 2-Ethyhexanoic acid is intended for use as a peptizer in the preparation of incendiary oil thickeners.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of unit packing and packing required (see 5.1).
(c) Quantity required for level C unit packing (see 5.1.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding off method of ASTM E29.

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